



■ Light It Up!

Circuits — Copper Tape Card



Science + Craft

Circuits & electricity through hands-on making!



30 Minutes

Build a real glowing card step by step!



Beginner

Grade 1 — Ages 5–7 No prior knowledge needed!

Priya wanted to make something special for her grandmother. She made a beautiful Diwali card that glows! When she pressed the card, a tiny light lit up like magic! Today, YOU will make your own glowing card — and learn real electricity science while doing it!



■ What is Electricity?

Electricity is like a tiny road for energy! It travels through wires to make lights glow bright. Just like **water flows in a river**, electricity flows through wires to bring us light and power! In our card, electricity will flow through copper tape to light up an LED!



■ The Three Key Science Words



BATTERY — The Energy Box!

A battery gives power to make things work! It is like a little energy box that stores electricity inside. Our coin cell battery is small and round like a coin — it powers LED lights in Diwali diyas and fairy lights!



CONDUCTOR — Road for Electricity

A CONDUCTOR lets electricity flow through it — like a smooth road for tiny energy cars! Copper tape is a metal conductor. Metals are good conductors. INSULATOR = wall that stops electricity. Paper and plastic block electricity and keep us safe!



CIRCUIT — The Complete Path

Electricity needs a COMPLETE path called a circuit. Think of it like a circle — energy travels all the way around! Like Diwali garland bulbs — if one wire breaks, ALL lights go off! The path must be complete for electricity to flow.



■ LED and Switch

■ LED

A tiny light bulb! You see LEDs in street lights and fairy lights. It has TWO legs: • Long leg (+) = the HAPPY side • Short leg (-) = the other side Always connect long leg to + side of battery!

■ SWITCH

A switch OPENS and CLOSES the circuit. It is like your bedroom light switch! • Press ON → electricity flows → light GLOWS • Press OFF → electricity stops → light OUT In our card, pressing the card folds the copper tape together to close the circuit!

■ Materials — What We Need



Copper Tape

Sticky metal tape — the road for electricity. 1 roll per group.



LED Light (1 per card)

Tiny light bulb with long (+) and short (-) leg. Handle gently!



Coin Cell Battery (CR2032)

Small round battery. Do NOT put in mouth. Adult supervision required.



Cardstock / Thick Paper

White or coloured — for folding and decorating your greeting card.



Scissors, Crayons, Stickers

For cutting and decorating. Make your card sparkle and shine!

■■ SAFETY NOTE: The coin cell battery (CR2032) is DANGEROUS if swallowed. Keep away from mouths. Adults must handle battery placement. Never short-circuit — do not touch both battery terminals at once.



■ Building Our Circuit Card — Step by Step

1

Fold Your Card

Fold your cardstock in half like a greeting card. Press down on the fold to make it nice and flat. This will be your glowing card!

2

Lay Copper Tape Path

Stick copper tape **INSIDE** your card to make a path for electricity. Press it down firmly so it sticks well. This is the **ROAD** for electricity! Leave a small gap — this will be your switch!

3

Insert LED Legs

Push the **LONG** leg (+) of the LED into one copper tape path. Push the **SHORT** leg (-) into the other tape path. Make sure legs touch the tape firmly! Long = positive, Short = negative — always!

4

Place the Battery

Slide the coin cell battery under the copper tape. The **SHINY** side (+) goes **UP**! Press gently so tape touches the battery. The battery completes the circuit!

5

Test Your Circuit!

Close the card (fold it). The copper tape should connect across the gap — this closes your switch! The LED should **GLOW**. If not, check steps below.

6

Decorate Your Card

Draw with crayons around your LED light. Add stickers to make it sparkle and shine — stars, flowers, Diwali diyas! Your LED light will shine through your art!



■ Test and Fix Your Circuit

If your light does NOT glow — don't worry! Every great scientist troubleshoots!

- Check copper tape is stuck down FLAT and FIRM all the way along.
- Check LED legs are TOUCHING the tape — press them down with your finger.
- Check battery shiny side (+) is FACING UP under the tape.
- Try pressing the card gently — sometimes the circuit needs a firm touch.
- Check the circuit is COMPLETE — no gaps except where your switch is.

■ Conductor vs Insulator

■ CONDUCTOR

Road for electricity! • Copper tape lets electricity flow • Like a smooth road for tiny energy cars • Metals are good conductors • Examples: copper, iron, gold, silver • Wires in your home are copper inside!

■ INSULATOR

Wall that STOPS electricity! • Paper and plastic block electricity • Like a wall that stops the energy cars • Keeps us safe from electricity • Examples: rubber, plastic, wood, glass • Why wires have plastic coating outside!



■ Reflect: What Did We Learn?

Q1

What makes the LED light glow?

Answer: A complete circuit — battery power flows through copper tape to the LED!

Q2

What happens if the battery lifts up?

Answer: The circuit BREAKS and the light goes OFF — electricity needs a complete path!

Q3

Name three parts of the circuit.

Answer: Battery (power source), copper tape (conductor path), LED (light)!

■ Session Outline — 30 Minutes

STORY & WARM-UP 5 MINS

Priya's Diwali card story.
Explore materials. Build
curiosity!

CONCEPTS 8 MINS

Electricity, battery,
conductor, insulator, circuit,
LED explained.

BUILD CARD 12 MINS

Follow 6 steps to build real
circuit card with copper tape
and LED.

TEST & FIX 3 MINS

Troubleshoot circuits.
Check tape, LED legs,
battery placement.

DECORATE 5 MINS

Crayons, stickers, draw
around LED. Make the card
sparkle!

REFLECTION 2 MINS

Answer 3 questions. Share
glowing cards. Celebrate!



■ Success Criteria



Every student successfully lights up their LED at least once.

- Students can name: battery, copper tape, and LED as circuit parts.
- Students explain in simple terms why the light goes off when the circuit breaks.
- Students used the words CONDUCTOR and CIRCUIT at least once each.
- No safety incidents — no child touched battery terminals with bare metal.
- Every child decorated their card and takes it home as a gift.



■ Extension Ideas

- Add a SECOND LED light to make your card extra bright and sparkly!
- Draw a NIGHT SCENE with your LED as a glowing moon or twinkling star!
- Make a PAPER FLAP SWITCH to turn your light on and off with a fold!
- Give your glowing card as a GIFT — for Diwali, a birthday, or just because!

Congratulations! You Made a Glowing Card! ■ Well done!
You made electricity work! Show your glowing card with
pride! Keep exploring and creating!